

BIM Guard User Manual

Architectural compliance checking for IFC models at bim-guard.xyz

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1 1. Introduction

BIM Guard checks IFC building models against architectural building-code rules, tracks the results as BCF issues and reports, and keeps your project documents under ISO 19650 governance. This manual walks through every page of the app at <https://bim-guard.xyz>, in the order you will normally use them.

Typical workflow

1. Create a project and attach an IFC model.
2. Upload a code document (PDF/text) and extract rules from it, or pick an existing ruleset.
3. Run a compliance audit.
4. Review results, 3D view, reports and exports (BCF, PDF, Excel, CSV).

1.1 Signing in

Open <https://bim-guard.xyz> and sign in with Google, or with email and password (a sign-up option is on the same screen). After sign-in you land on the Dashboard.

1.2 Screen layout

- Left sidebar – navigation. At organization level it shows My Home, Rules & Standards and Coordination & Graph; inside a project it switches to project tools (Dashboard, Models, Run Compliance Audit, Reports & Exports, Evaluation, Graph-RAG Copilot, 3D Viewer, Query Console, Live Pipeline). All Projects returns you to the organization level.
- Organization switcher (top of sidebar) – change the active organization.
- Top bar – breadcrumb, Documentation and Integrations menus, Admin portal, and a light/dark theme toggle.
- Resources (bottom of sidebar) – Modeling Manual, User Manual, Revit IFC Export Setting, bSDD Wiki.

2 2. Projects

2.1 2.1 Dashboard (project registry)

The dashboard lists every project with its status, whether an IFC model is attached, analysis domain, jurisdiction and creation date. Use the filter box and dropdowns to narrow the list, click a column header to sort, and use the row buttons to open a project's 3D view or audit. Manage Repos connects a model storage source.

2.2 2.2 New Project

A six-step wizard: Details → IFC Model → Naming → Scope → Inputs → Confirm.

1. Details – client name, project name, short name, project code, description, jurisdiction and project type. Fields marked * are required, and some unlock only after the client name is entered.
2. Continue through the steps to attach the IFC model, set ISO 19650 naming, scope and inputs, then confirm.

2.3 2.3 Project Dashboard

Opening a project shows its domain and status plus cards for Models, Compliance Audit, 3D Viewer, Reports & Exports and Live Pipeline.

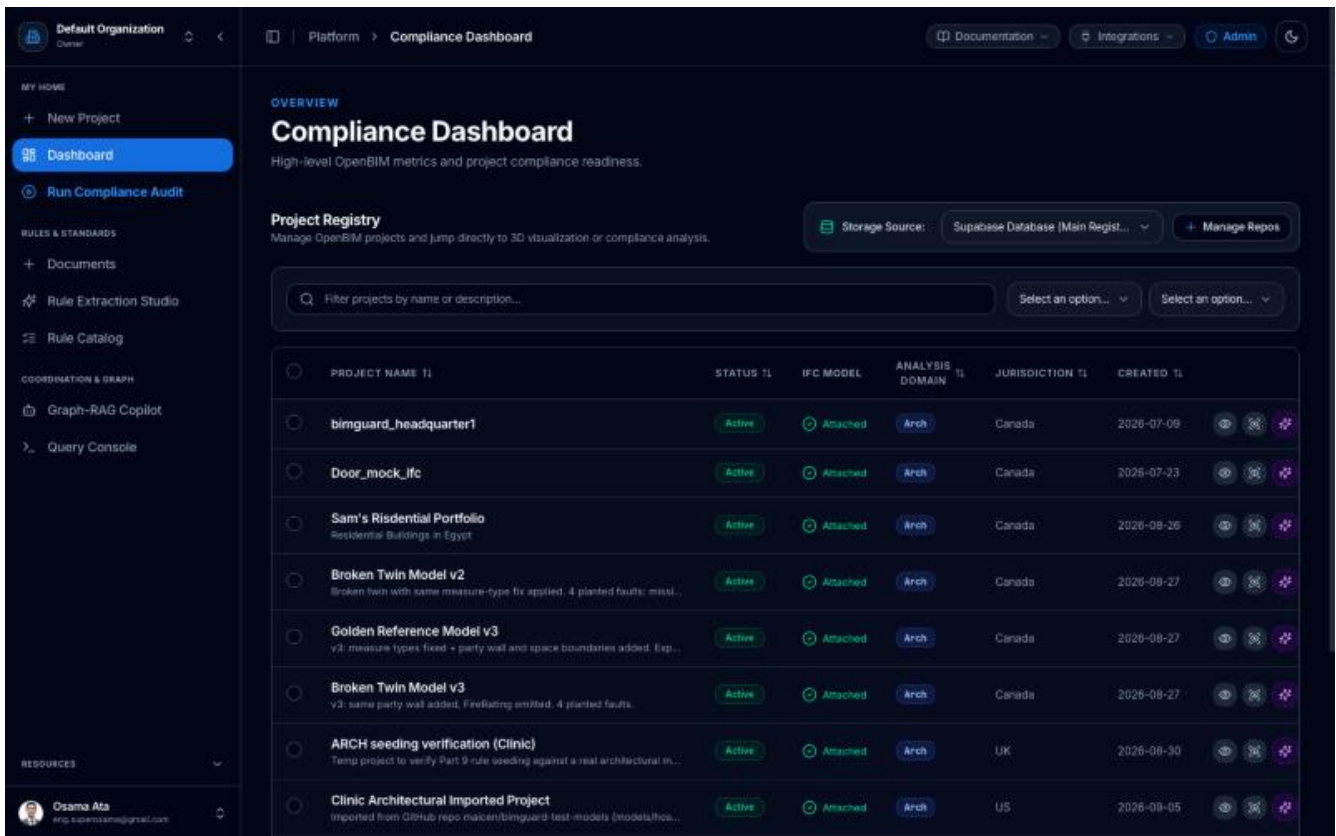


Figure 1: Compliance Dashboard

2.4 2.4 Models

Lists the IFC models attached to the project. Use Attach Model to add the primary model or supporting context files. A project with no model shows an empty state with the same button. Projects need a model before the audit or 3D viewer can be used.

2.5 2.5 3D Viewer

Shows the attached model with a docked panel for Properties, BCF Topics, Spatial Hierarchy and Layers. Select an element to inspect its properties; open a BCF topic to jump to its viewpoint. The screenshot shows the viewer while the graphics engine loads on a project that has no model attached yet.

3 3. Documents and rules

3.1 3.1 Documents

The document library holds building codes, specifications and project manuals.

- Upload Specification – add a PDF or text file.
- Add Manual Rules, OpenCDE Sync, Import from Drive – other ways to bring content in.
- The DocLang column shows whether a structured conversion is Ready or can be started with Convert.
- Search, type filter, sortable columns, row actions and pagination (10/25/50/100 per page) work as on every table in the app.

3.2 3.2 Rule Extraction Studio

Turns natural-language code clauses into machine-executable rules.

Figure 2: New Project Setup

1. Pick a Source Specification Document (or Add / Upload Document).
2. Choose the extraction model/parser.
3. Click Extract Compliance Rules, then review and approve the proposed rules. Your edits to a proposed rule are recorded, so corrections stay traceable.

3.3 Rules Catalog

Browse and manage every rule. Rulesets (folders) are on the left; the table shows rule reference, category, mechanism (for example CODE) and target property. Search by id, description or property, filter by mechanism, or show only rules that Need Review. A reliability legend (High / Medium / Low) tells you how dependable the checked property is. Top-right actions: Seed Rules, Import / Export, IDS / JSON Ruleset, Manual and + New Rule.

3.4 Manual Rule Editor

Write rules by hand. Choose the ruleset folder to save into (or create one with New Folder), pick a building-element category (Windows, Doors, Stairs, Ramps ...) and click Add Rule against a known property. Press Save Changes when done.

3.5 bSDD Wiki

Read-only browser for the local buildingSMART Data Dictionary cache — the same classes and properties that power the rule builder's autocomplete. Filter by dictionary, entity, property set or quantity set, then pick a class to see its definition and standard properties.

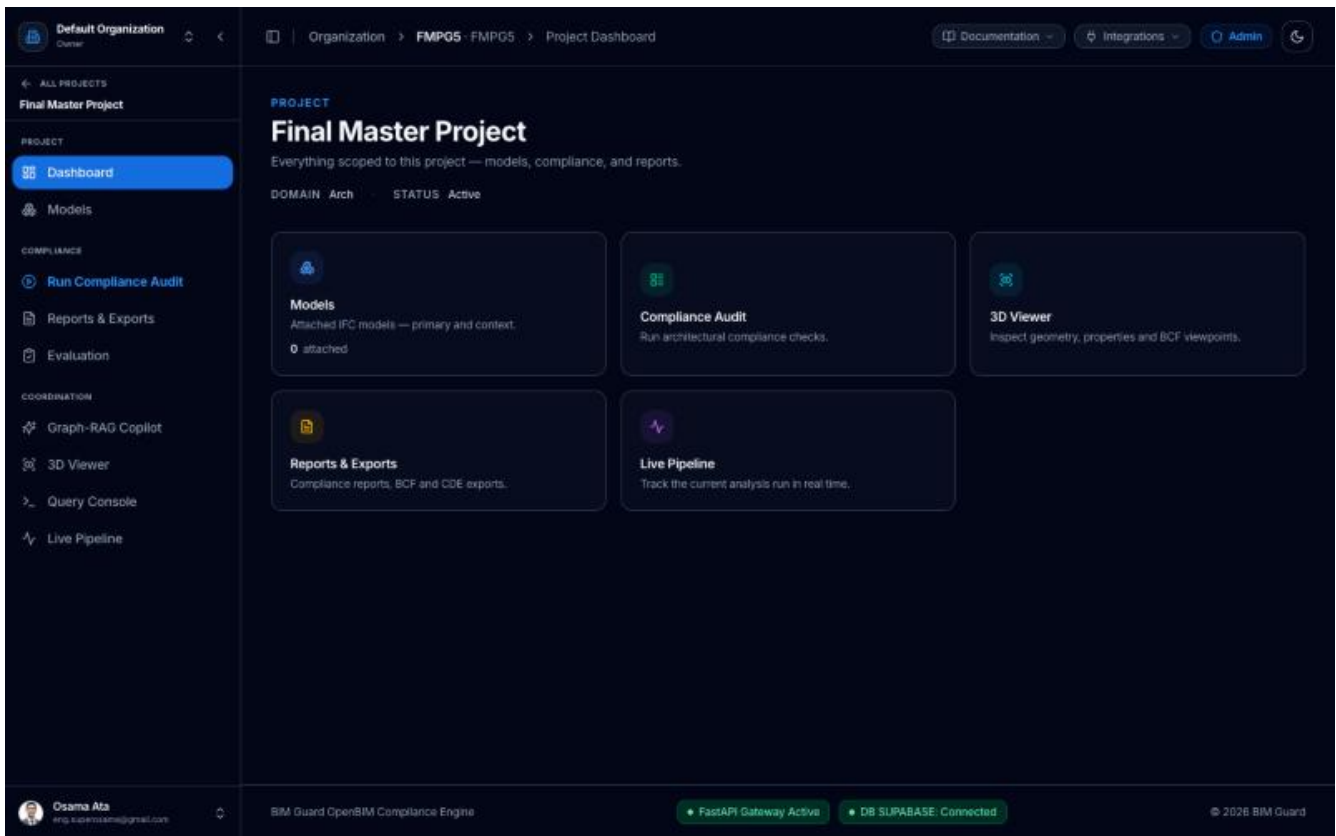


Figure 3: Project Dashboard

3.6 3.6 Document Viewer

Open a document from the library to read it. The left pane shows the Original Page and the right pane the cleaned Reading View; Export .dclx saves the structured DocLang version, and Views switches the layout. Rule-Source Map jumps to the rules extracted from this document.

3.7 3.7 Rule-Source Map

Shows which rules were extracted from a document and which model elements they map to. The Rules tab lists approved rules; Pending Drafts lists extracted rules awaiting review. A ruleset-level Source Map (reached from the Rules Catalog) shows which documents a whole ruleset came from.

4 4. Running and reviewing compliance

4.1 4.1 Run Compliance Audit (quick start)

A guided three-step flow from the sidebar:

1. Choose a project – existing or create new.
2. Pick a ruleset – an existing one, or upload a document and extract rules.
3. Run the audit.

4.2 4.2 Architectural Compliance Audit

Inside a project, select a Ruleset (or All Rules) and press Run Compliance Audit (shown as Attach IFC Model to Audit when no model is attached yet). Results appear in the panel below.

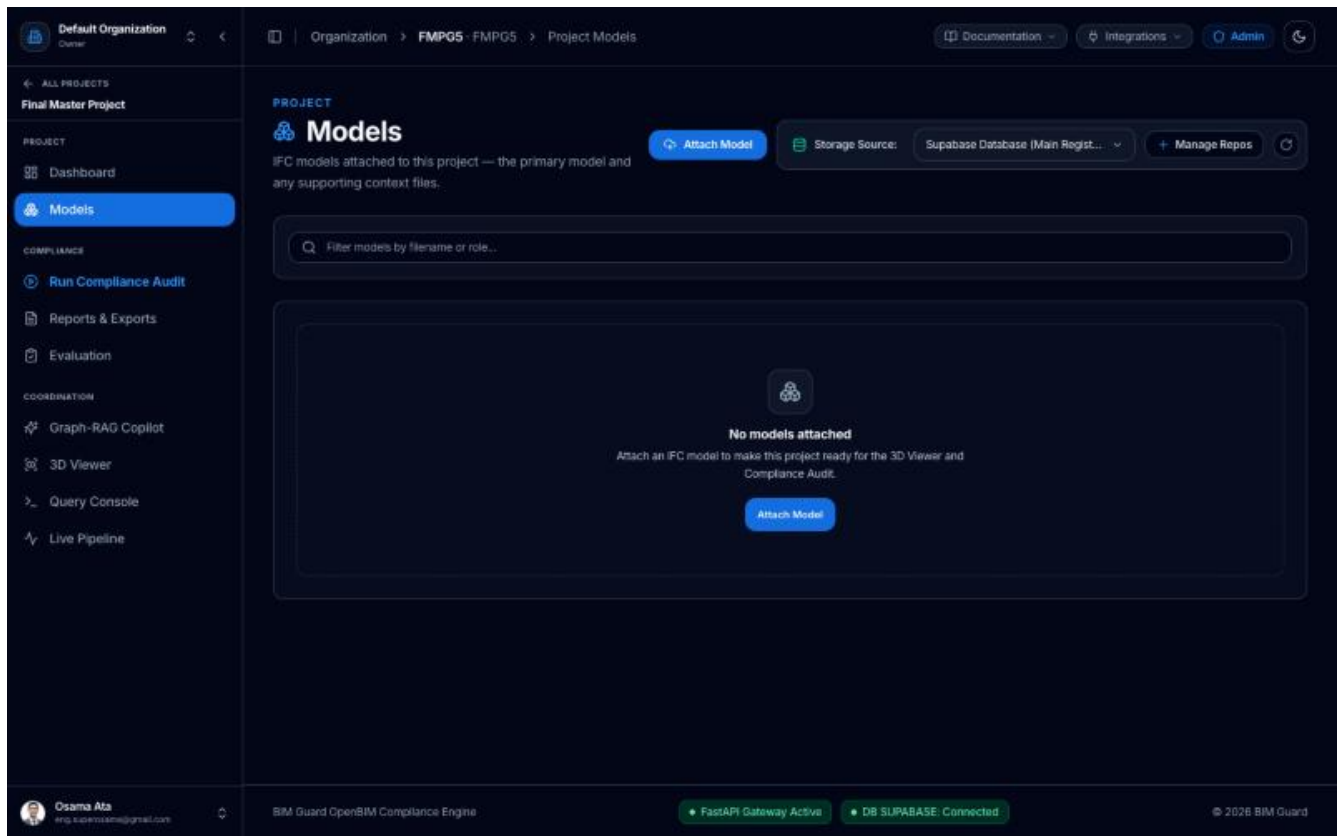


Figure 4: Project Models

4.3 4.3 Live Pipeline

Real-time progress of the run via server-sent events across six stages: Validation, IFC Parsing, Engine Execution, Risk Scoring, Report Assembly and Export. The Engine Execution Matrix shows each engine's state (Pending, Running, Done).

4.4 4.4 Reports & Exports

- BCF Collaboration Hub – Live BCF 2.1 Topics (issues exchanged over the BCF API) and BCF Zip Files saved from each audit.
- Compliance Report – Preview or Download PDF; also Excel and CSV exports.
- Export W3C PROV-O (.ttl) – provenance of the analysis.

4.5 4.5 Evaluation

Confirm or override BIM Guard's PASS/FAIL verdicts after an audit (use Capture for Evaluation). Agreement is scored as a confusion matrix with precision, recall and F1.

5 5. Graph-RAG and queries

5.1 5.1 Graph-RAG Copilot

Ask questions about documents, the IFC model, or both, with answers grounded in the project's knowledge graph and cited. Pick the project at top right and the Knowledge Scope (Hybrid, Document, Model). Chats are saved in the left column and searchable. Suggested follow-ups give quick starting questions.

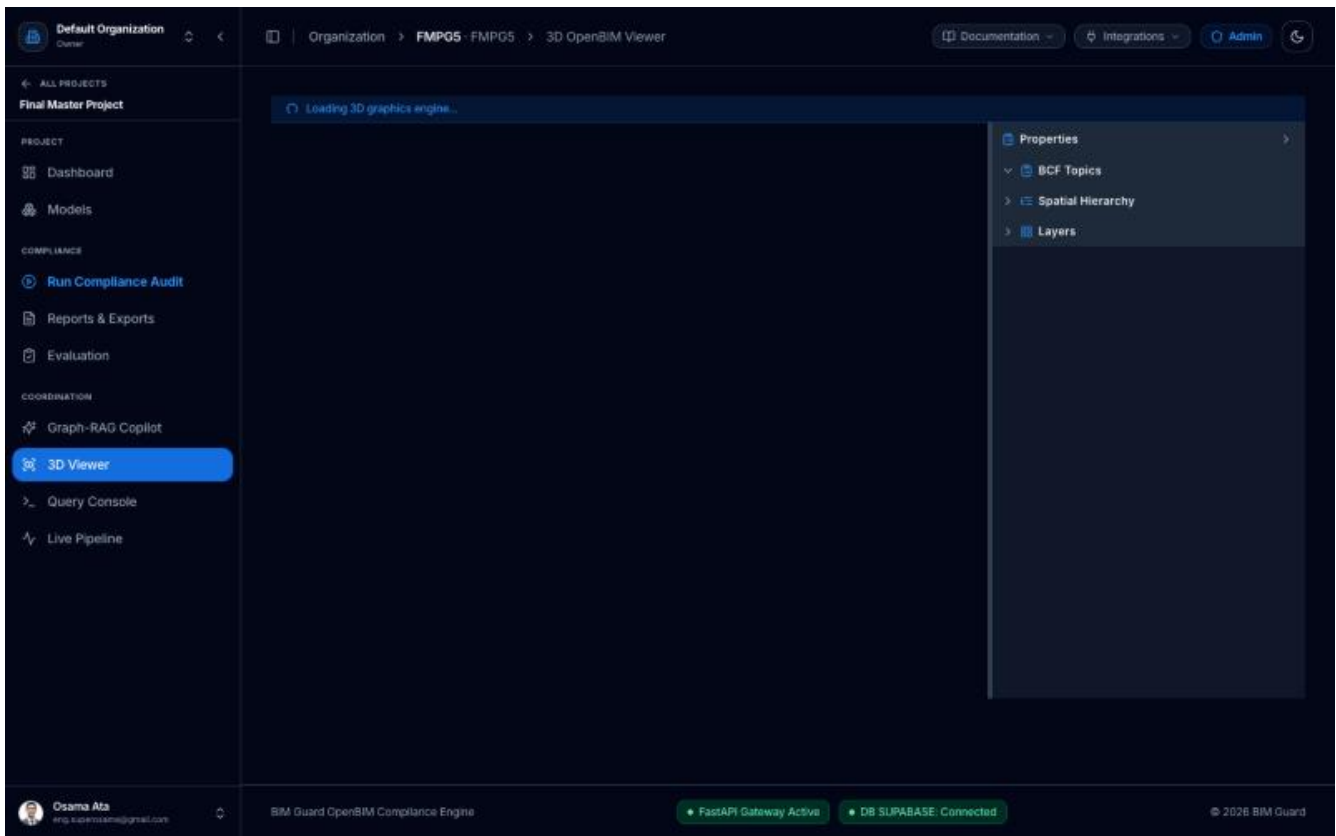


Figure 5: 3D OpenBIM Viewer

5.2 Query Console

Tabs: Graph-RAG Copilot, Model Health, Regulatory GraphRAG, Cypher Presets, SPARQL Endpoint and Code-to-BIM Trace — for advanced inspection of the property graph.

6. Integrations and resources

6.1 Revit Direct Sync

Audit directly from Revit without exporting IFC. Copy the pyRevit / IronPython push script into a pyRevit button; it POSTs element data to the gateway endpoint shown. Simulate Push tests it.

6.2 IFC Export Setting

Download a preconfigured IFC4 export profile (.json) and a setup guide for Revit so exports contain the property sets, quantities and GUIDs BIM Guard needs. Check that the exported file starts with `FILE_SCHEMA(('IFC4'))`.

6.3 User Workflow Manual

The in-app step-by-step guide: create a project and upload the IFC, inspect it in the 3D viewer, add reference documents, and so on. Each step has a button that jumps straight to the right page.

6.4 3D Modeling Reference

Modeling conventions that make an IFC pass automated checks — for example doors must be true `IfcDoor` elements, hosted in a wall, assigned to a storey and connected to room-bounding spaces — plus egress, daylight and

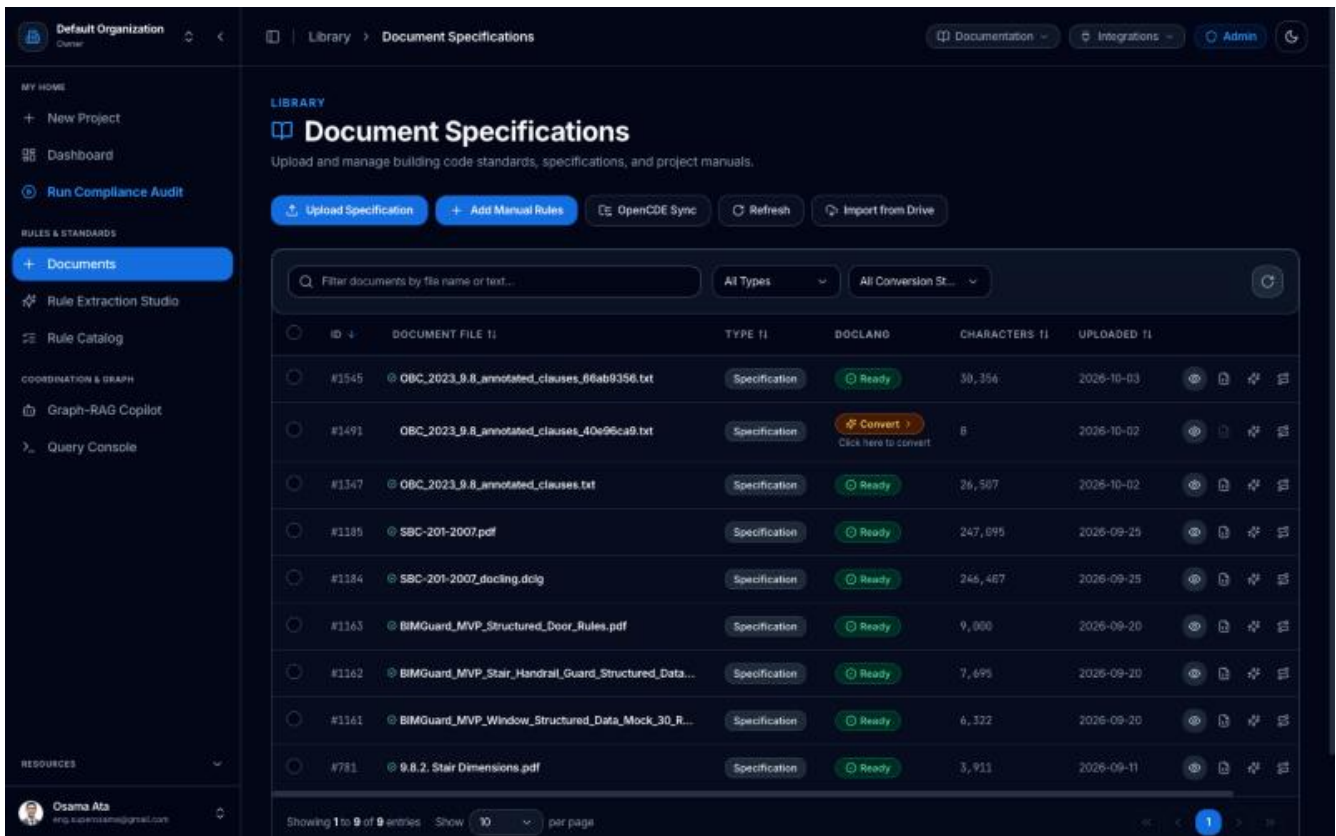


Figure 6: Document Specifications

stair/handrail guidance.

7. Settings and administration

7.1 Runtime Settings

Edit your display name and title, and choose Dark, Light or System Auto appearance. Persistence backend and log level are shown for reference; Save Settings stores changes.

7.2 Admin Portal

Reach it with the Admin button (owners/admins). Its sidebar contains: Organization settings, External providers, Role permissions, Users & organizations, Ruleset access, Project access and Document access. Back to Main App returns you to the workspace.

7.2.1 Organization settings

Members and pending invites for the active organization. Change a member's Role or Group, Remove them, or add people with Invite member. Groups (below the table) control project visibility: a plain member sees only the projects their group is granted, while owners and admins see every project. Member names and emails in this manual's screenshots are anonymised.

7.2.2 External providers

Tabs for Document Parsing, LLM Providers and Environment. Add parsing engines for your organization (they are used in preference to the platform default), or review the platform defaults. Each instance can be tested (Test),

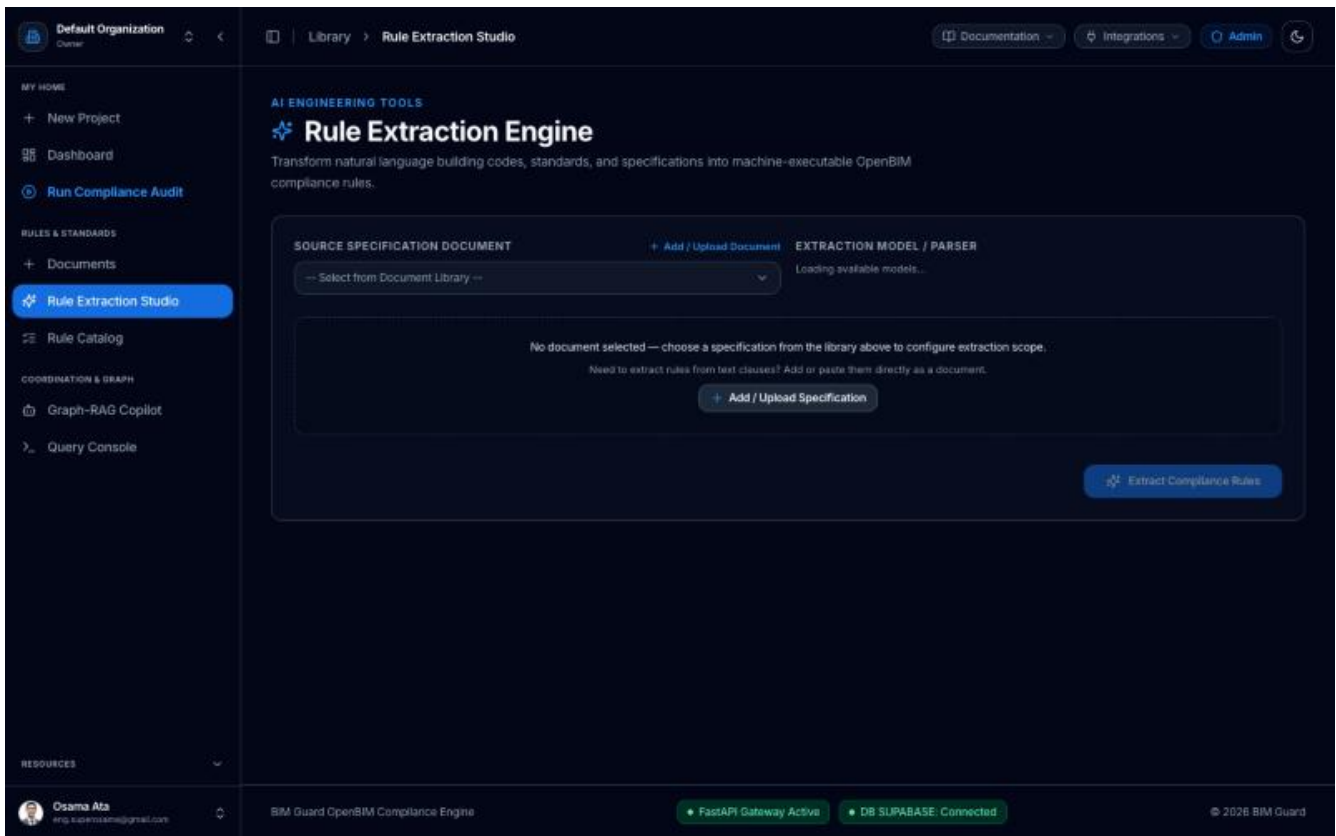


Figure 7: Rule Extraction Studio

made the default, or disabled. Endpoint URLs are masked in the screenshot.

7.2.3 Role permissions

Choose which role each gated action requires (for example `manage_org_members`, `manage_llm_providers`), as a platform default or a per-organization override.

7.2.4 Ruleset access

Grant or revoke each organization's access to rulesets so they can be bound to its projects.

7.2.5 Users & organizations

Platform-wide list of organizations (with slug, org code and member count) and users. Create an organization with New Organization, assign a user to organizations or remove them with the row actions, and search by email, name or organization. Superadmin only; user identities are anonymised in the screenshot.

7.2.6 Project access

A matrix of projects against organizations showing Granted, Owner or No Access, for sharing a project across tenant boundaries.

7.2.7 Document access

Controls which specification documents an organization may bind to its projects. Filter by type or status and grant or revoke per document.

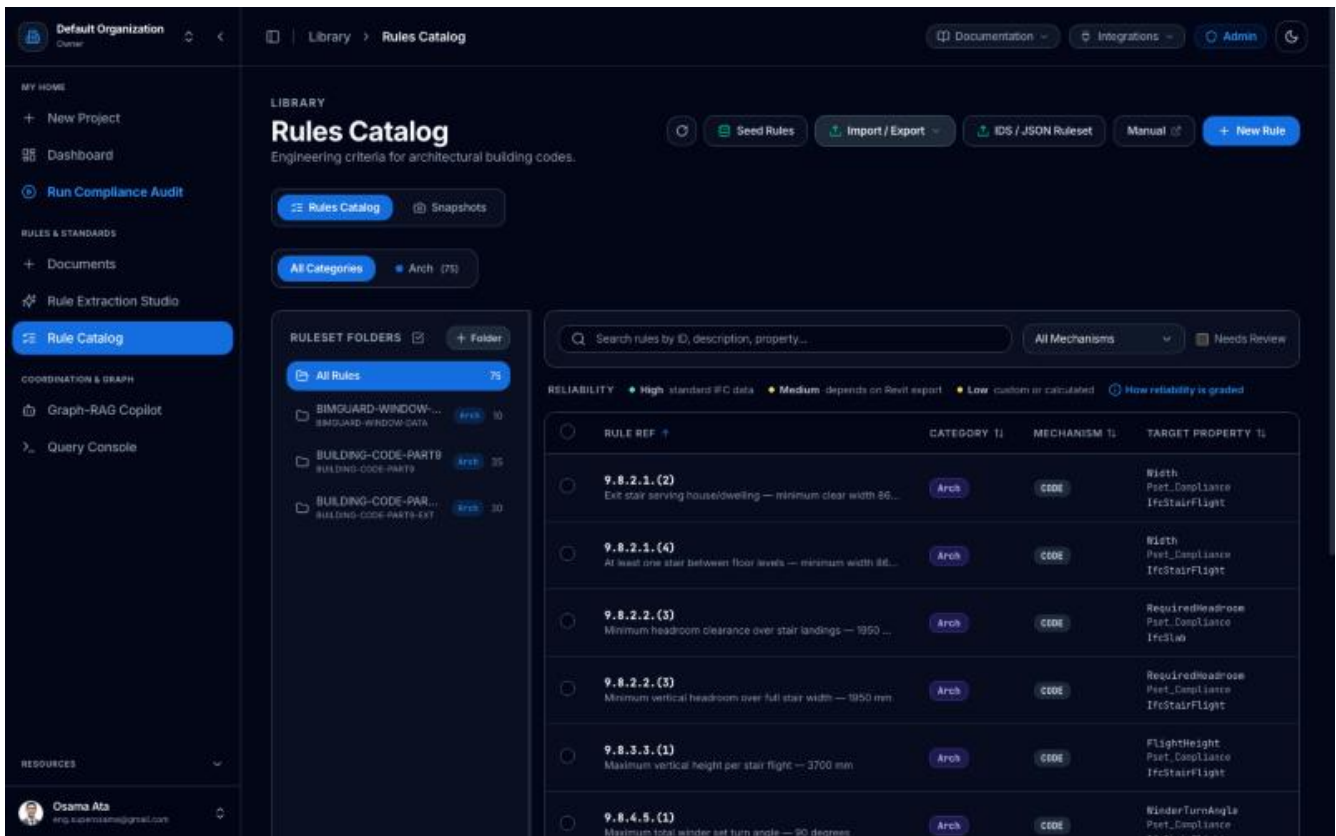


Figure 8: Rules Catalog

8. Tips

- Every table supports search, filters, column sorting, multi-row selection with bulk actions, and pagination.
- Project-specific pages carry `?project_id=` in the address, so you can bookmark or share a link to a project view.
- Not seeing a project? Ask an owner to add you to a group with access.

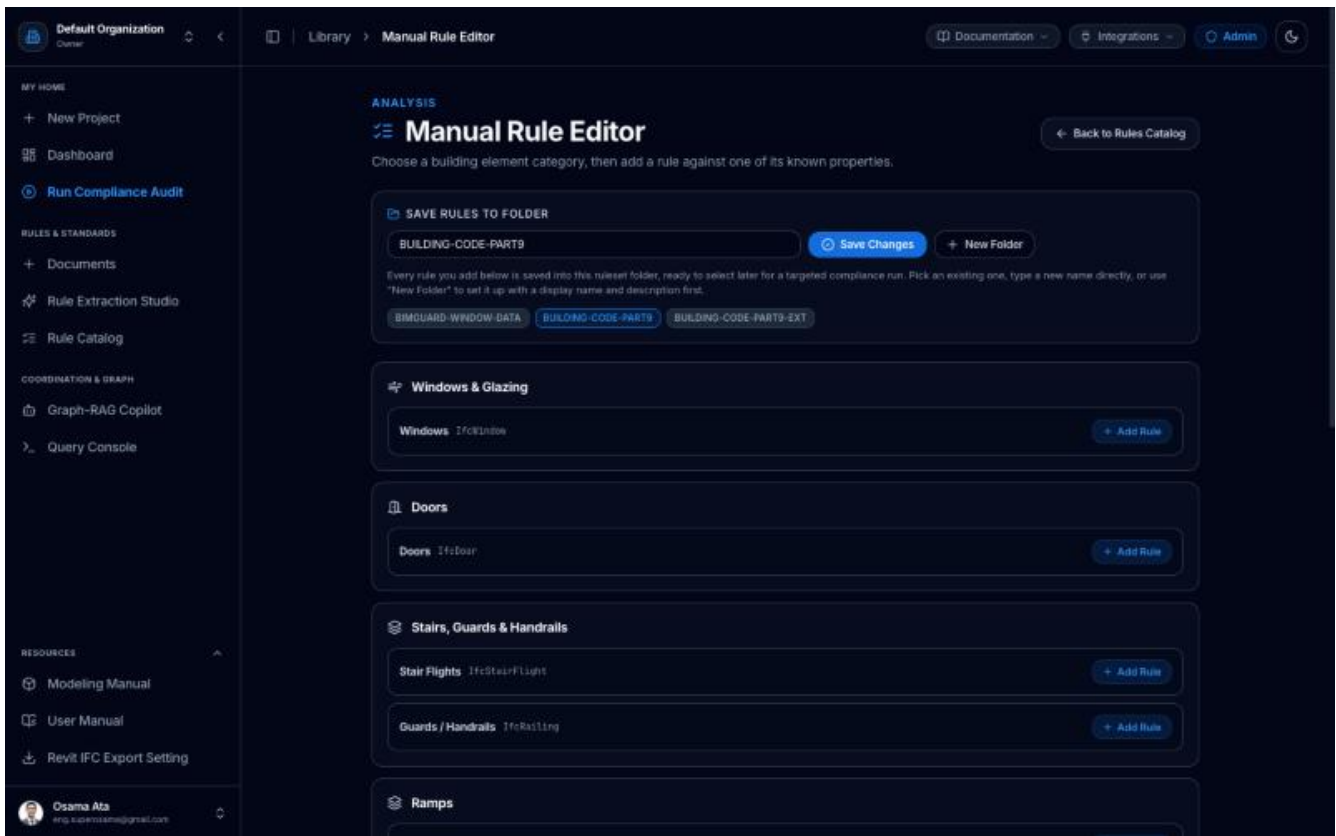


Figure 9: Manual Rule Editor

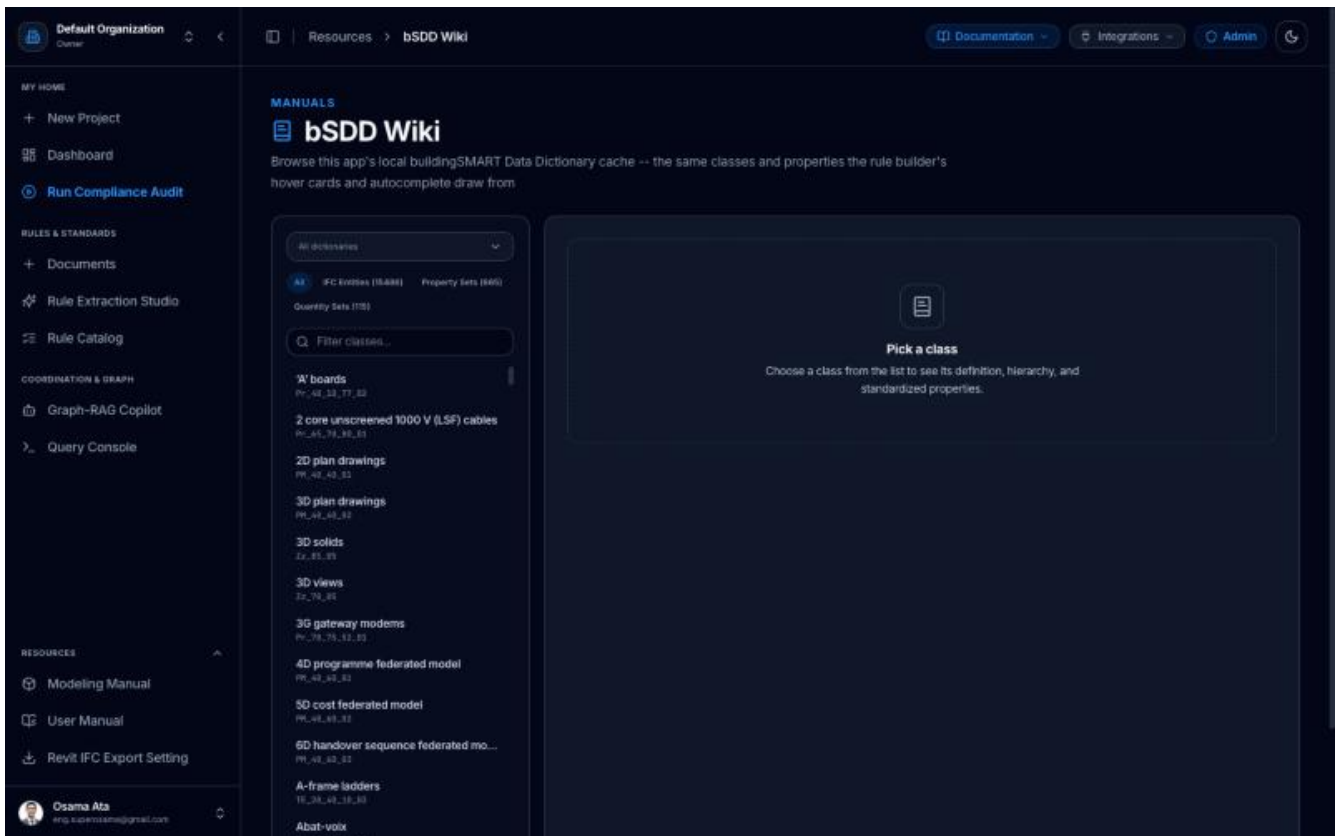


Figure 10: bSDD Wiki

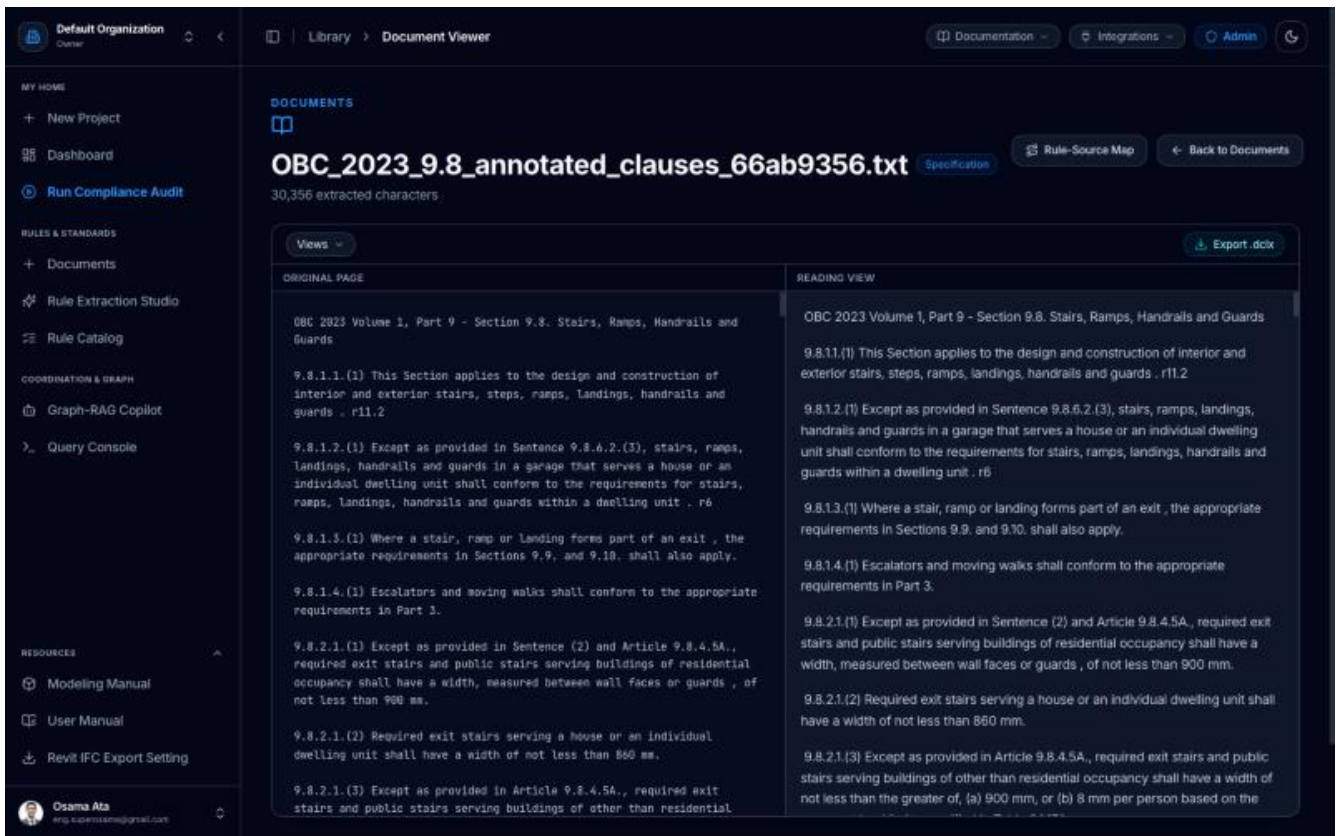


Figure 11: Document Viewer

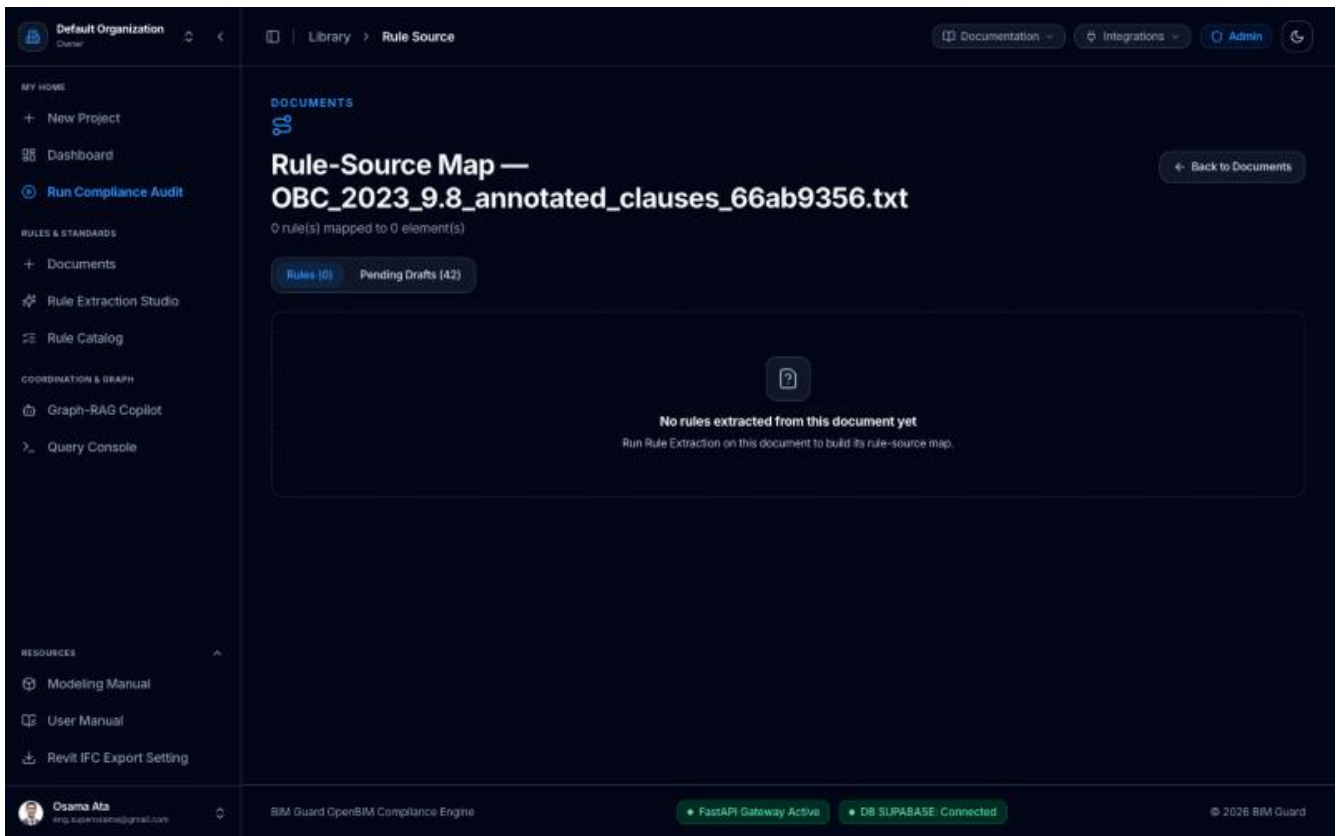


Figure 12: Rule-Source Map

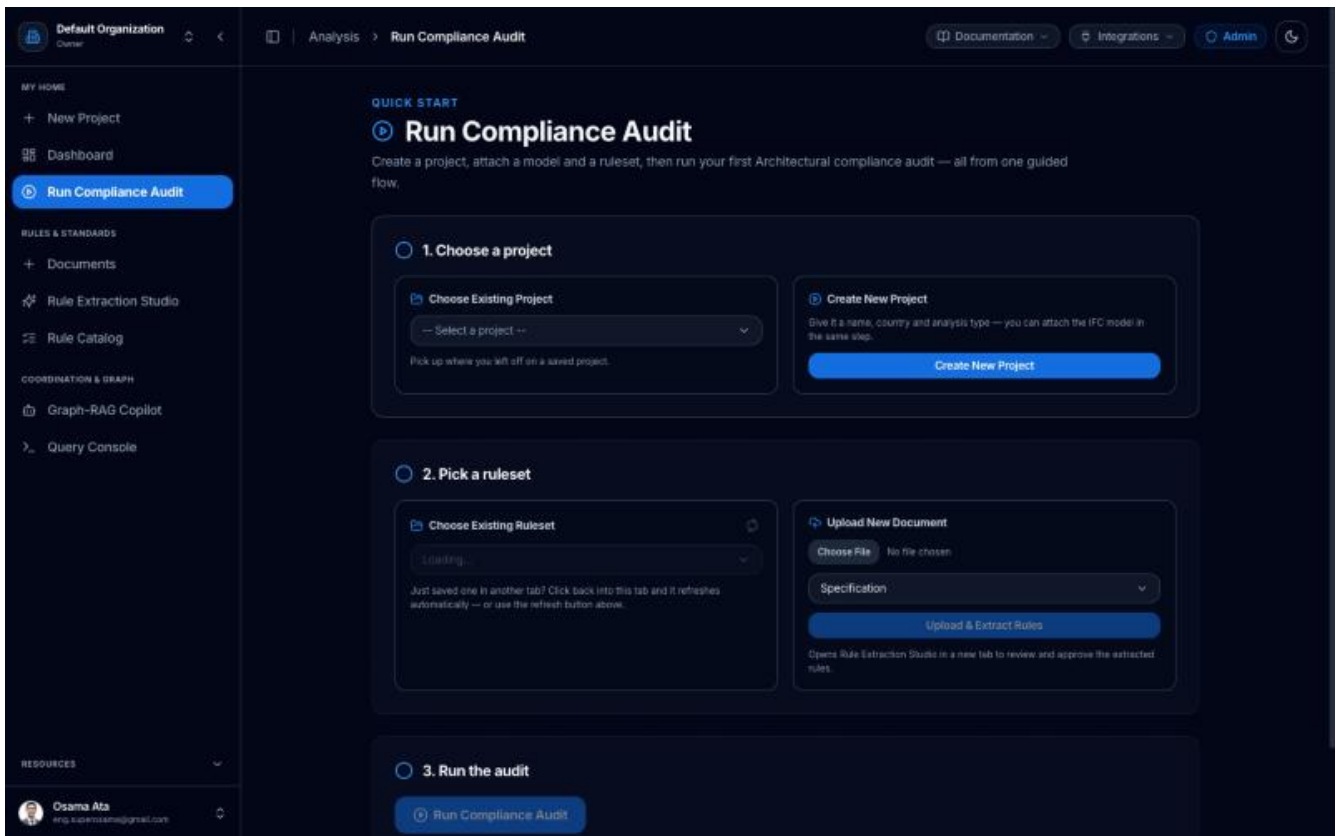


Figure 13: Run Compliance Audit

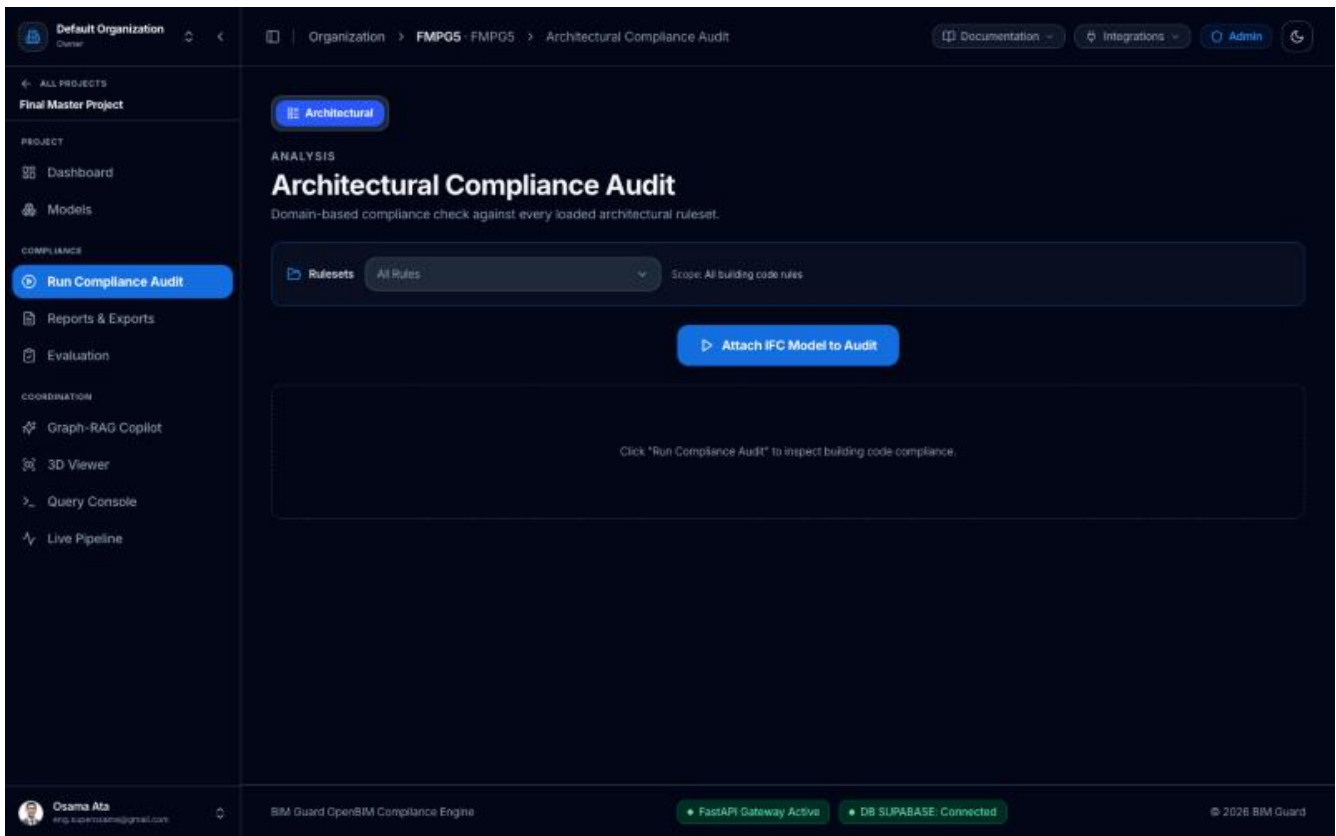


Figure 14: Architectural Compliance Audit

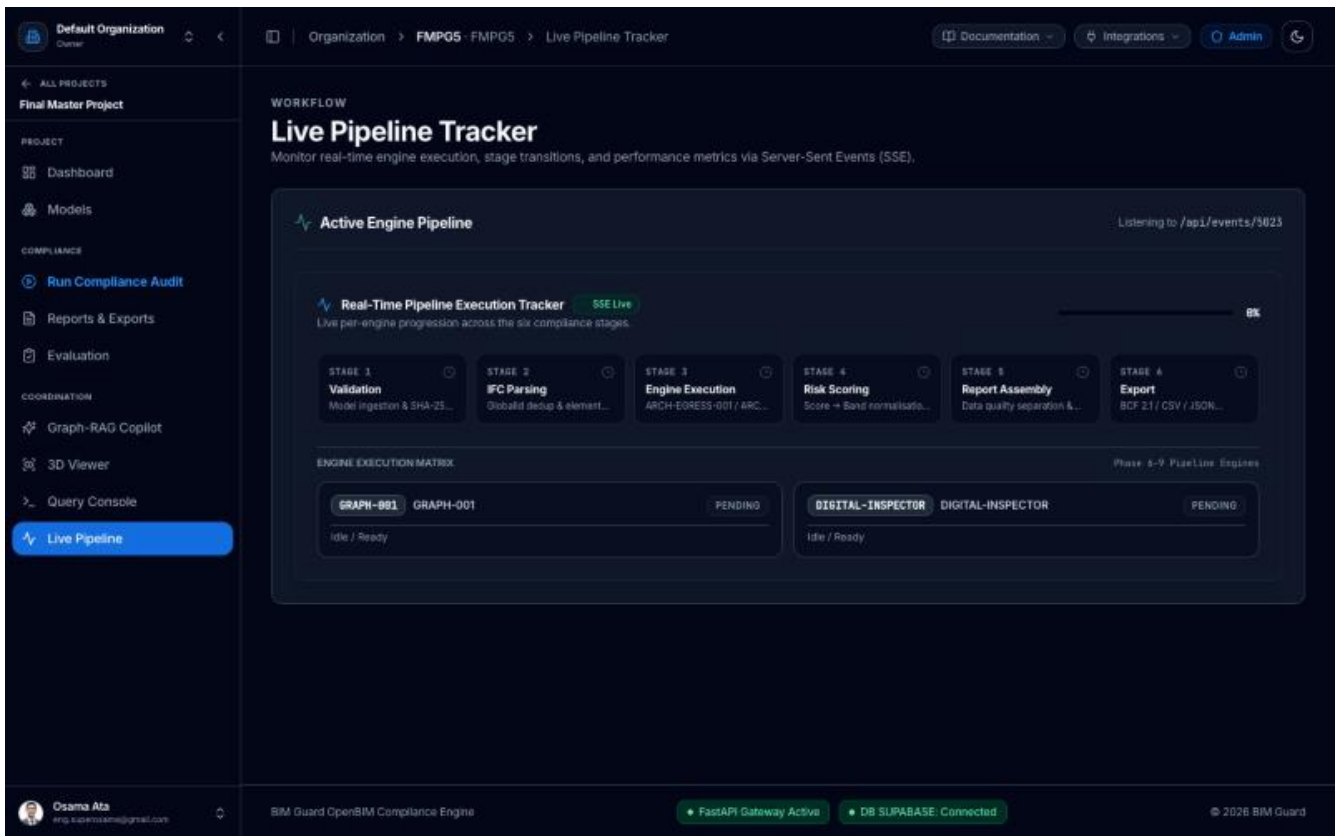


Figure 15: Live Pipeline Tracker

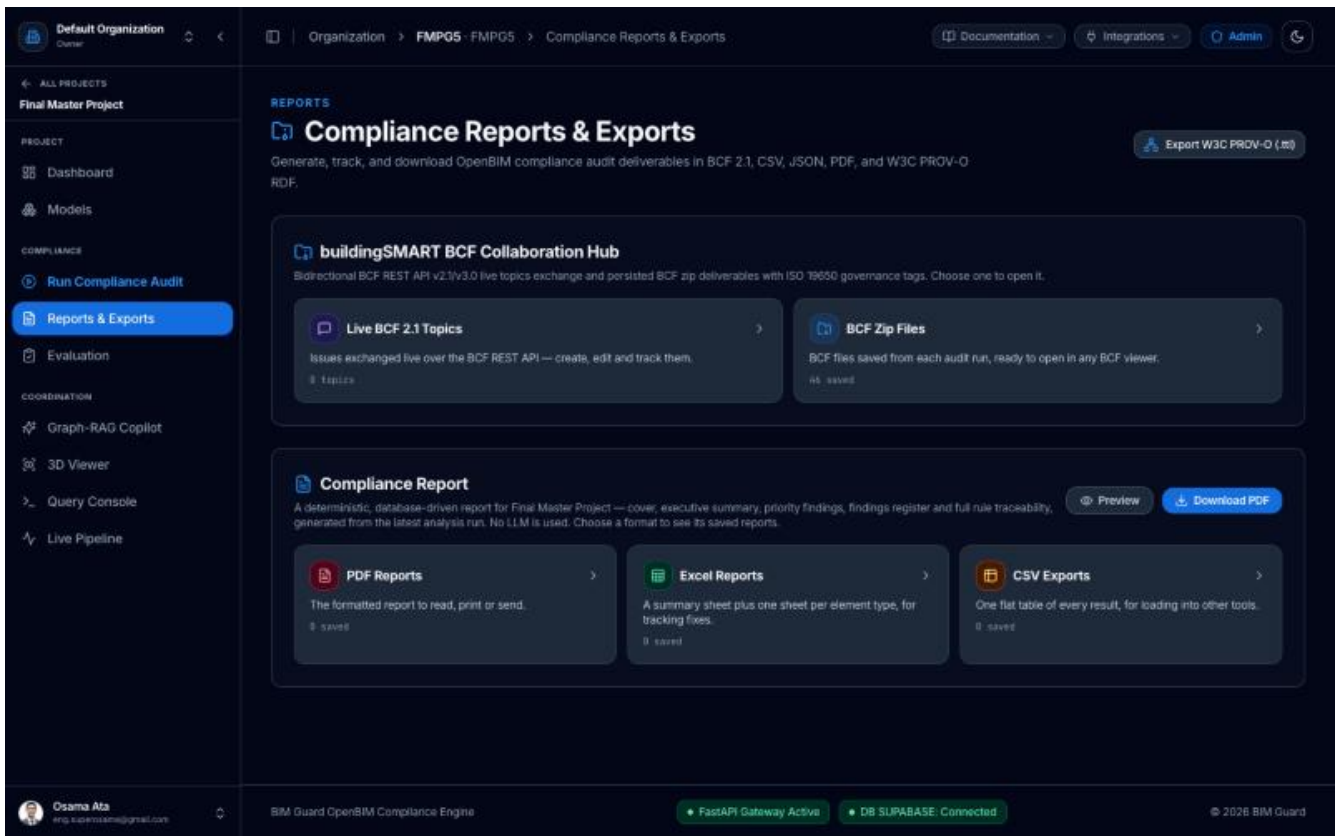


Figure 16: Compliance Reports & Exports

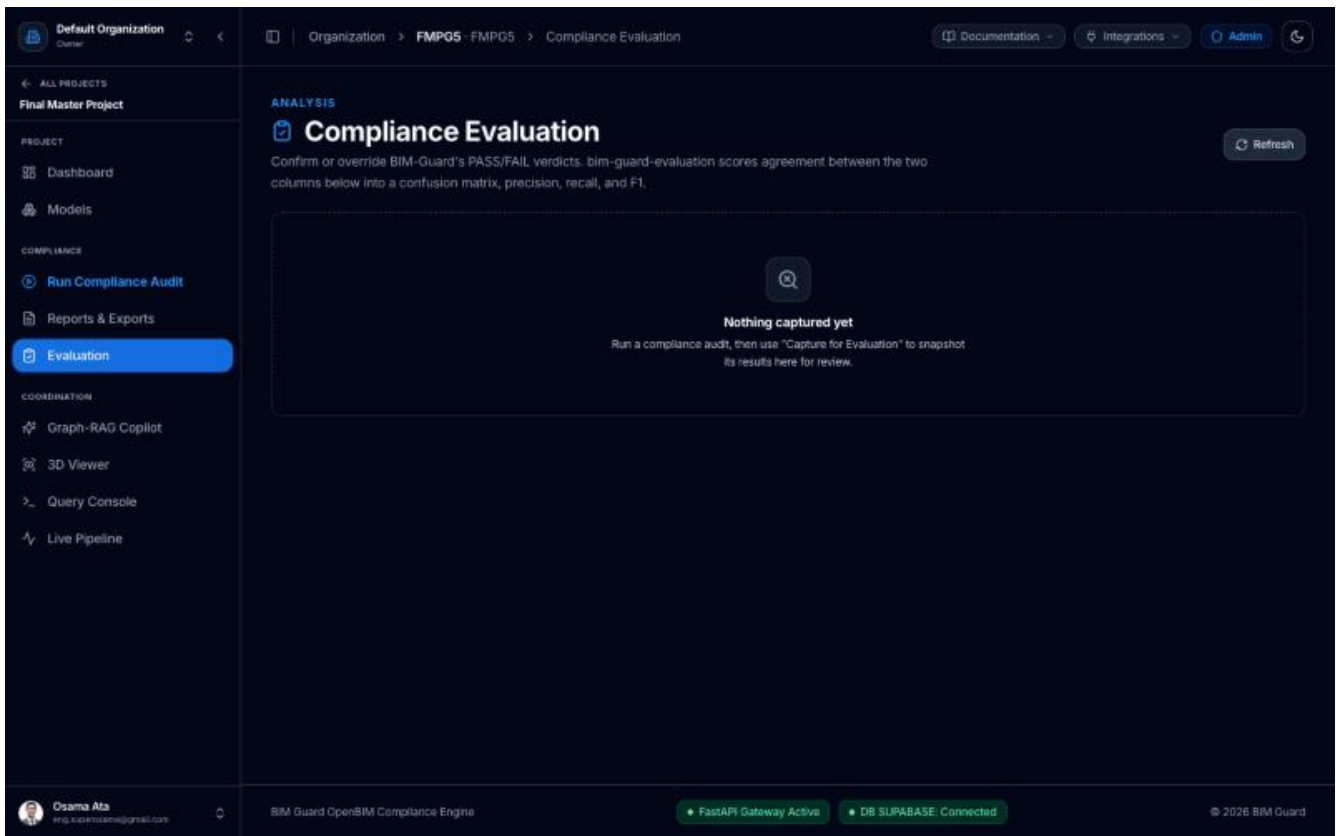


Figure 17: Compliance Evaluation

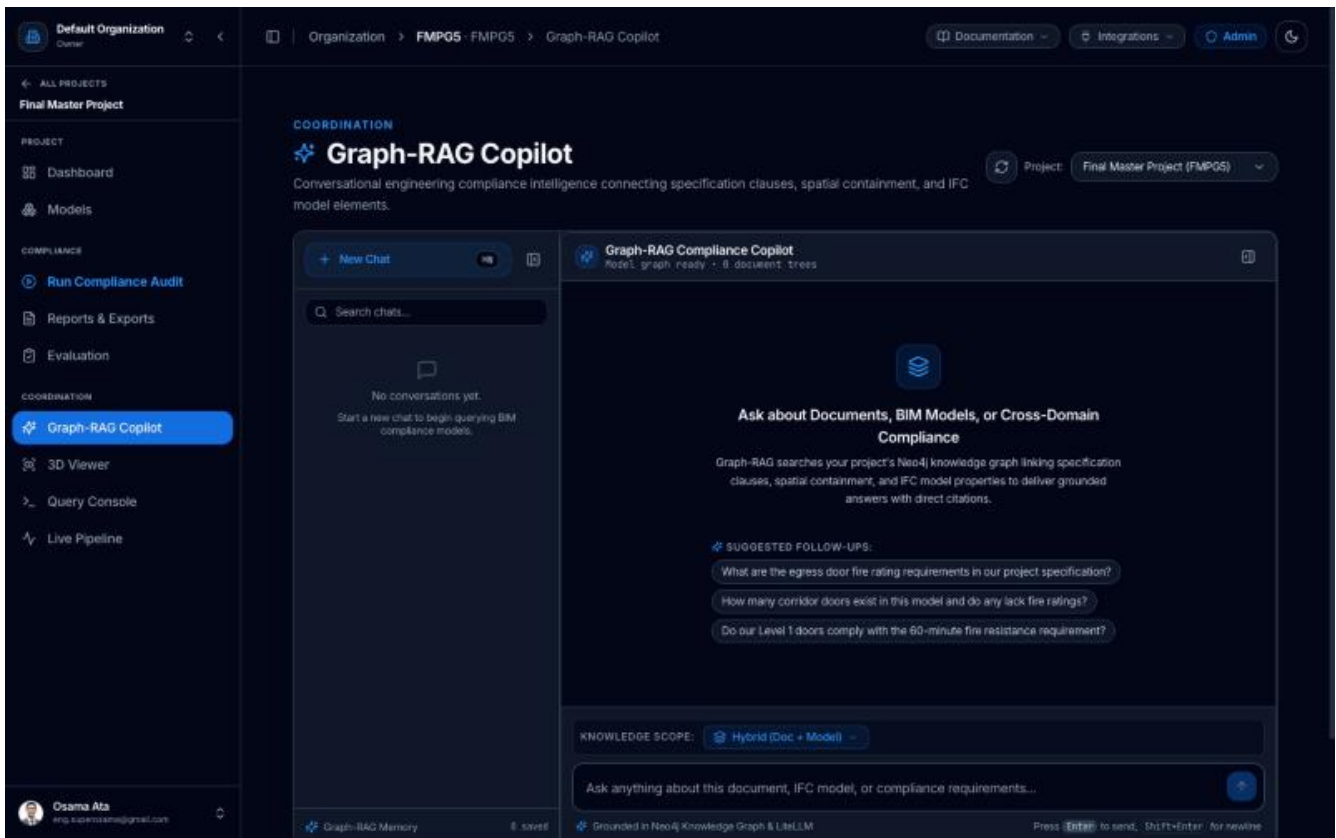


Figure 18: Graph-RAG Copilot

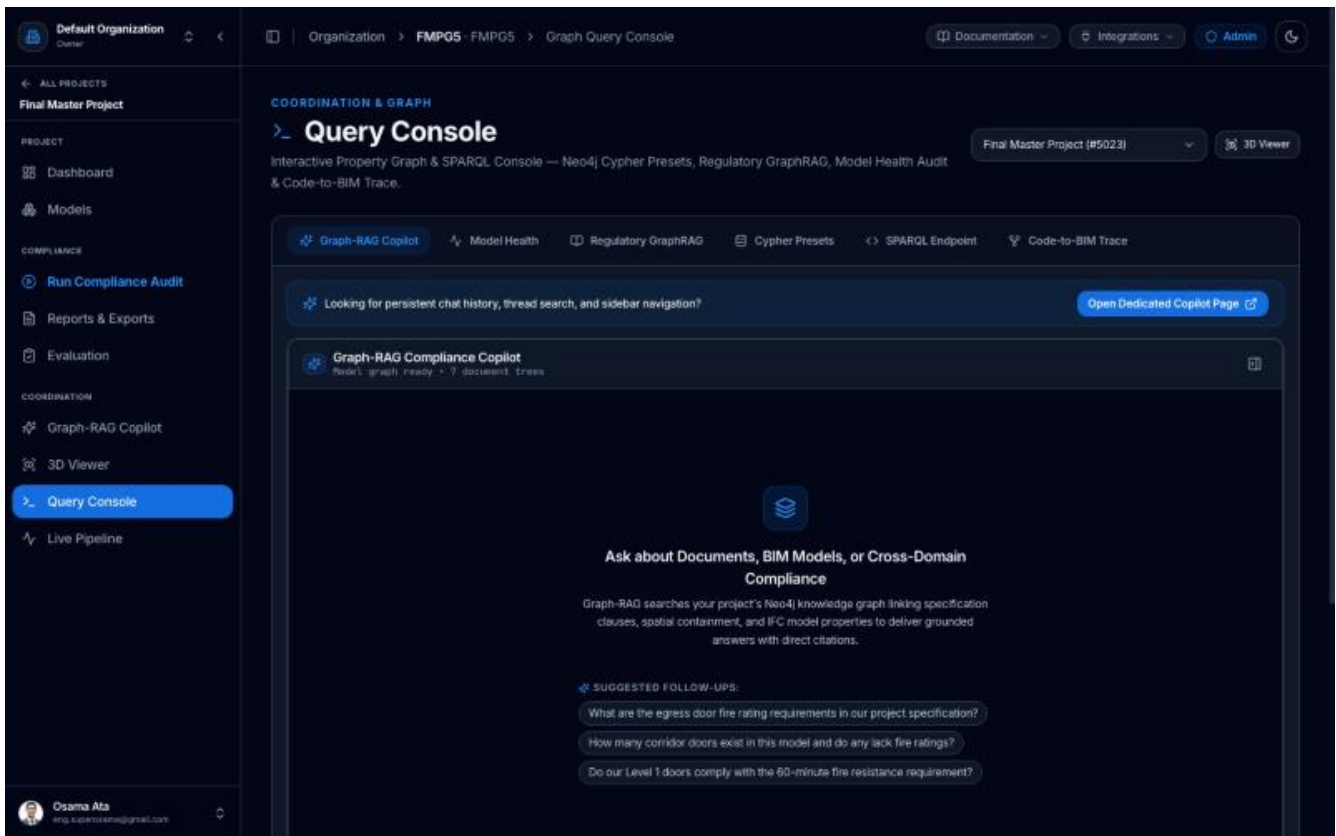


Figure 19: Query Console

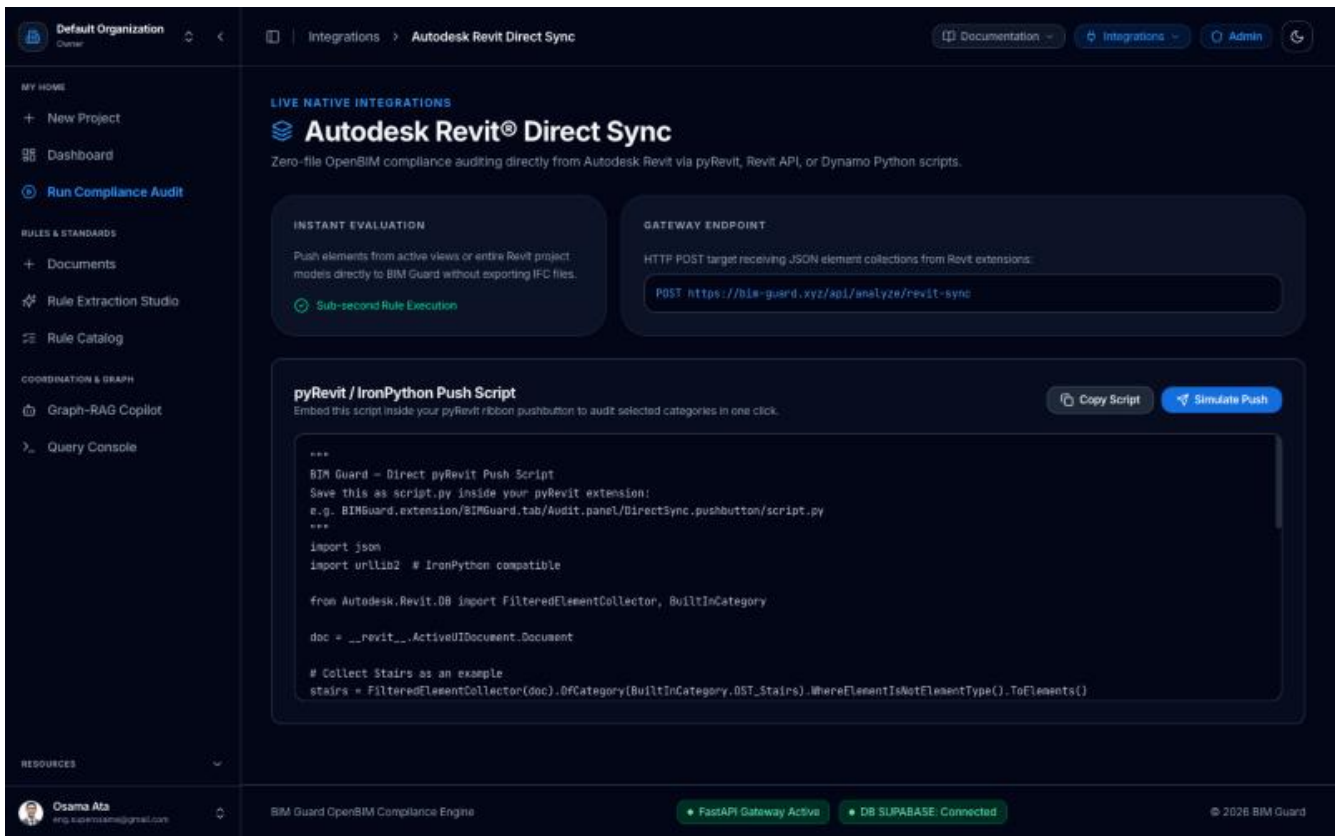


Figure 20: Autodesk Revit Direct Sync

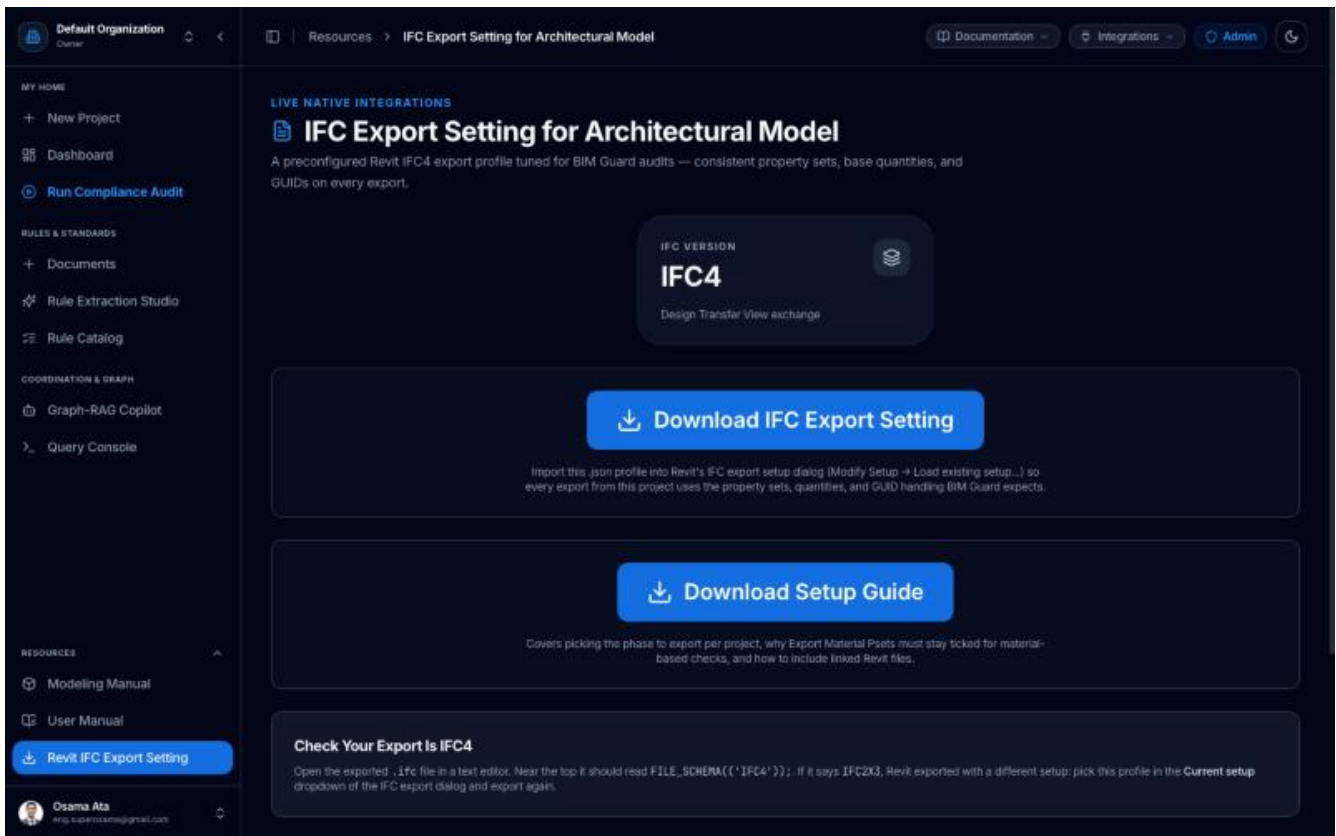


Figure 21: IFC Export Setting

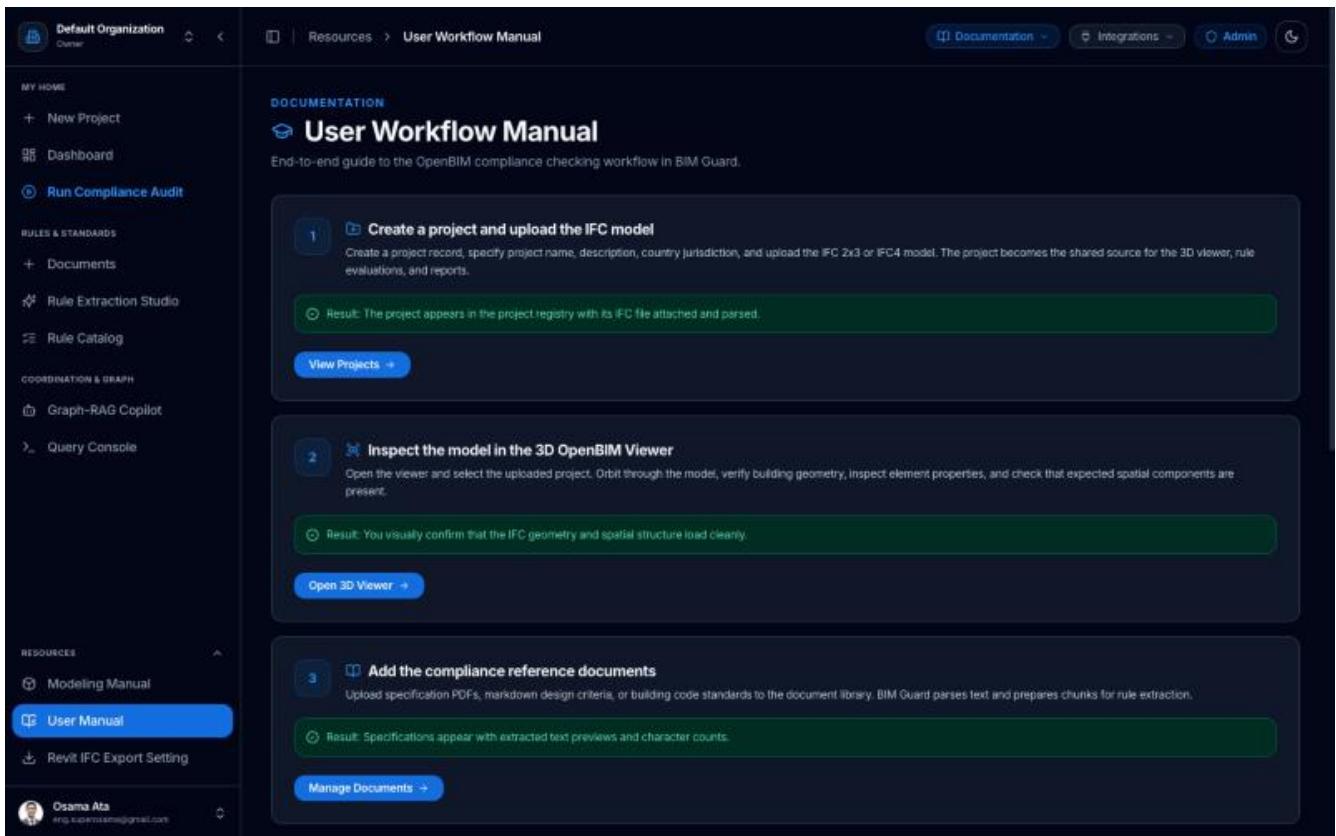


Figure 22: User Workflow Manual

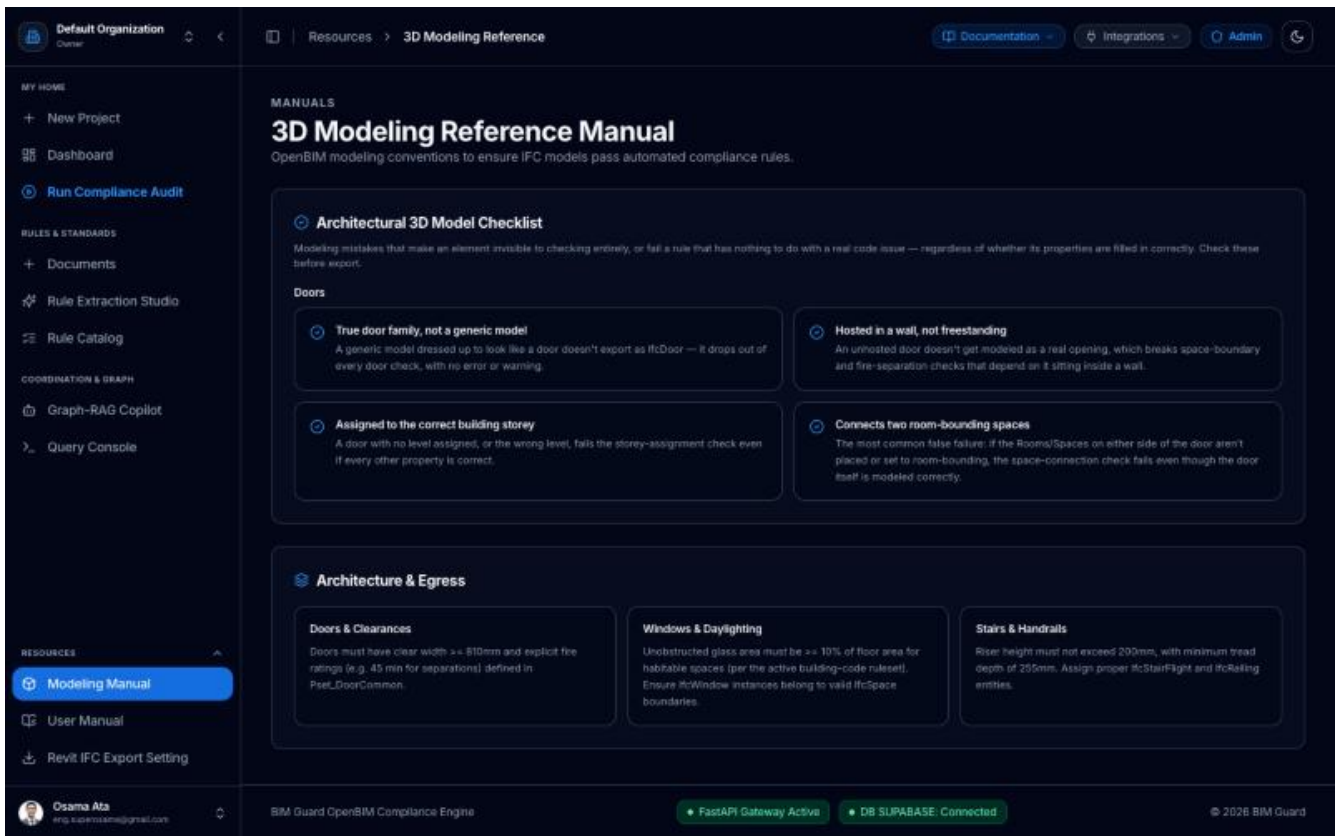


Figure 23: 3D Modeling Reference

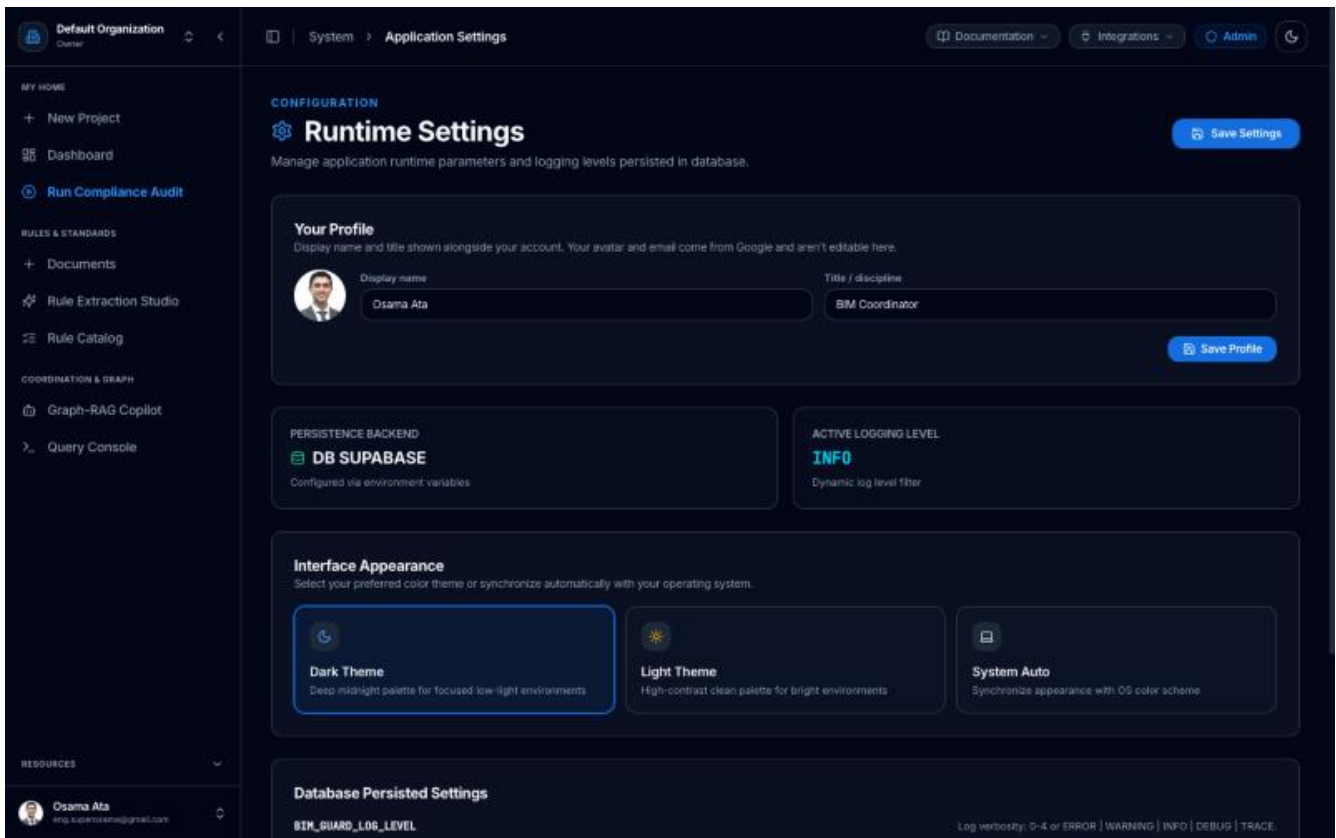


Figure 24: Runtime Settings

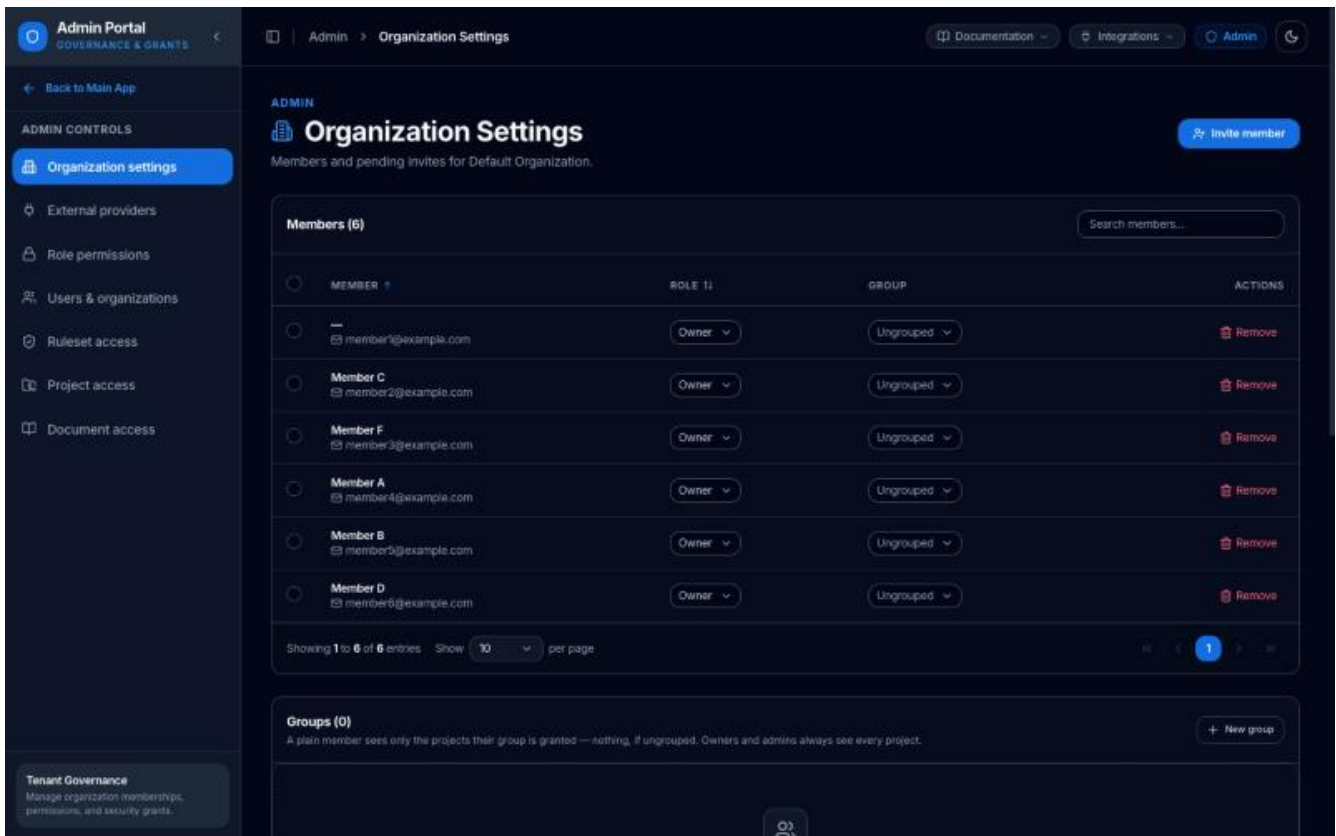


Figure 25: Organization Settings

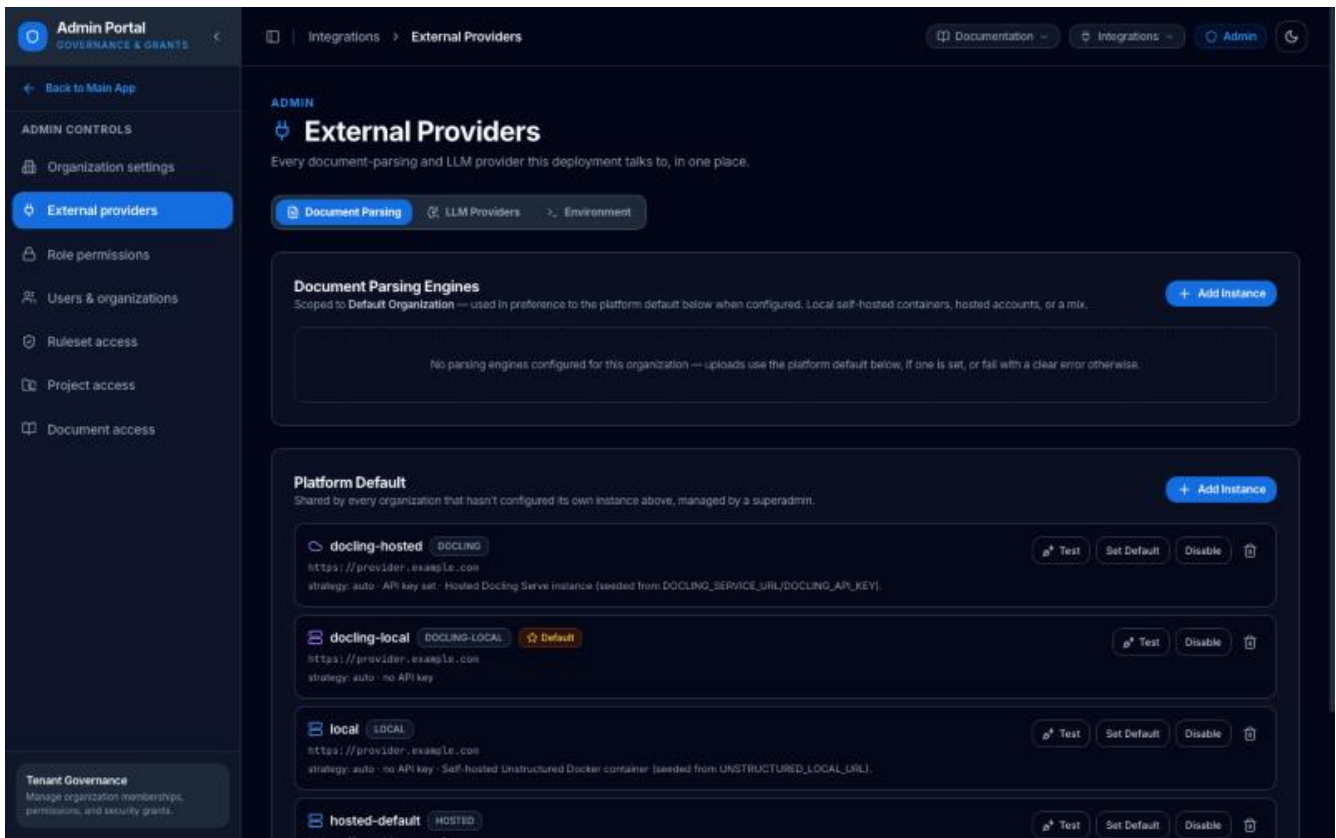


Figure 26: External Providers

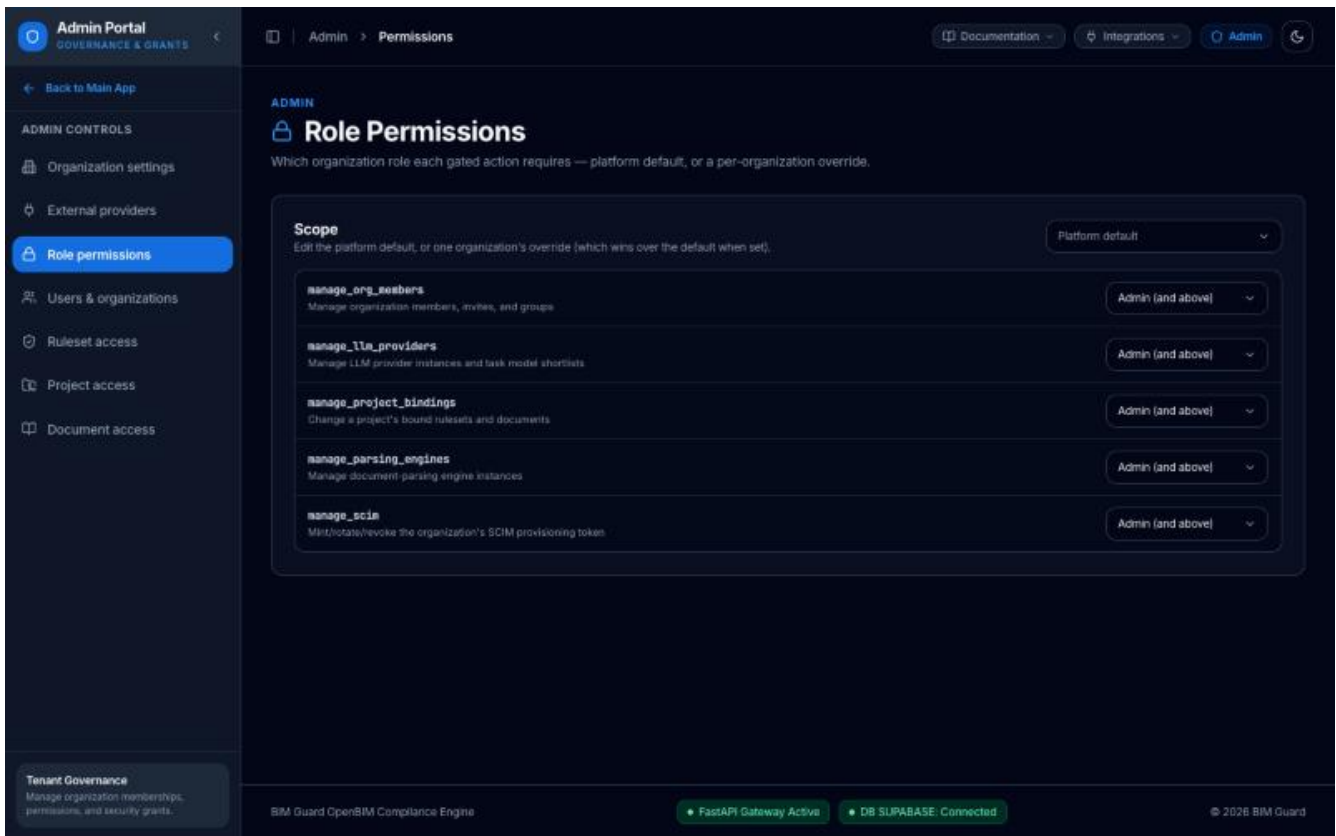


Figure 27: Role Permissions

Admin Portal
GOVERNANCE & GRANTS

Back to Main App

ADMIN CONTROLS

Organization settings

External providers

Role permissions

Users & organizations

Ruleset access

Project access

Document access

Tenant Governance

Manage organization memberships, permissions, and security grants.

Governance
Ruleset Access

Documentation
Integrations
Admin

PLATFORM GOVERNANCE

Ruleset Access

Manage which engineering compliance rulesets each organization is granted to bind to its projects.

Search rulesets by name, ID, or description...

Default Organization

All Categories

All Statuses

Reset

☐	RULESET NAME ↑	RULESET ID ↑	CATEGORY ↑	RULES ↑	Default Organization Access	Actions
☐	BIMGUARD-WINDOW-DATA	BIMGUARD-WINDOW-DATA	Arch	18	Granted	
☐	BUILDING-CODE-PART9	BUILDING-CODE-PART9	Arch	35	Granted	
☐	BUILDING-CODE-PART9-EXT	BUILDING-CODE-PART9-EXT	Arch	18	Granted	

Showing 1 to 3 of 3 entries
Show 10 per page

BIM Guard OpenBIM Compliance Engine

FastAPI Gateway Active
DB SUPABASE Connected

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Figure 28: Ruleset Access

Admin Portal

GOVERNANCE & GRANTS

← Back to Main App

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Organization settings

External providers

Role permissions

Users & organizations

Ruleset access

Project access

Document access

Tenant Governance

Manage organization memberships, permissions, and security grants.

Governance > User Access

Documentation

Integrations

Admin

PLATFORM GOVERNANCE

Users & Organizations

+ New Organization

See every user on the platform, assign or remove them from organizations, and manage organizations themselves.

ORGANIZATIONS (4)

Name	Slug	Org Code	Members	Delete
Awesome Developer	awesome-developer	AWD	2	
Aspiring Design 3D Consultancy Ltd	aspiring-design-3d	AD3D	3	
Archinova	archinova	ARCH	5	
Default Organization	default	BTNG	6	

USERS (6)

Search by email, name, or organization...

User	Organizations	Actions
Member A member1@example.com	Archinova - owner X Default Organization - owner X	
Member B member2@example.com	Archinova - admin X Default Organization - owner X	
Member F member3@example.com	Default Organization - owner X	
Member E member4@example.com	Default Organization - owner X Archinova - member X Aspiring Design 3D Consultancy Ltd - member X	
member5@example.com	Default Organization - owner X Archinova - owner X Aspiring Design 3D Consultancy Ltd - owner X Awesome Developer - admin X	
Member D member6@example.com		

Figure 29: Users and Organizations

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GOVERNANCE & GRANTS

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External providers

Role permissions

Users & organizations

Ruleset access

Project access

Document access

Tenant Governance

Manage organization memberships, permissions, and security grants.

Governance > Project Access

Documentation
Integrations
Admin

PLATFORM GOVERNANCE

Project Access

Cross-organization project sharing — which organizations may access projects across tenant boundaries.

Search projects by name, ID, country...

All Organizations (4)

All Access

PROJECT NAME	COUNTRY	Awesome Developer	Aspiring Design 3D Consultancy Ltd	Archinova	Default Org
ARCH seeding verification (Clinic) #83 · Arch	UK	No Access	Granted	No Access	De
B-GIT Test Project #3992 · Arch	Canada	No Access	No Access	No Access	De
bimguard_headquarter1 #8 · Arch	Canada	No Access	No Access	Granted	De
Broken Twin Model v2 #28 · Arch	Canada	No Access	No Access	Granted	De
Broken Twin Model v3 #22 · Arch	Canada	No Access	No Access	Granted	De
Clinic Architectural Imported Project #1985 · Arch	US	No Access	Granted	No Access	De
Door_mock_lfc #11 · Arch	Canada	No Access	No Access	Granted	De
Final Master Project #5823 · Arch	Canada	No Access	No Access	No Access	De
Gate1 Blocked Project #5282 · Arch	GB	No Access	No Access	No Access	De
Gate1 Clean Project	GB	No Access	No Access	No Access	De

Figure 30: Project Access

Admin Portal

GOVERNANCE & GRANTS

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Project access

Document access

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Manage organization memberships, permissions, and security grants.

Governance

Document Access

Documentation

Integrations

Admin

PLATFORM GOVERNANCE

Document Access

Which specification documents each organization is permitted to bind to its projects.

Search documents by filename, code, originator...

Default Organization

All Types

All Statuses

Reset

DOCUMENT NAME ↑	TYPE ↑	Default Organization Access	Actions
9.8.2. Stair Dimensions.pdf #781 (RIP)	Specification	Granted	
BIMGuard_MVP_Stair_Handrail_Guard_Structured_Data_Mock_3D_Rules.pdf #1142 (RIP)	Specification	Granted	
BIMGuard_MVP_Structured_Door_Rules.pdf #1143 (RIP)	Specification	Granted	
BIMGuard_MVP_Window_Structured_Data_Mock_3D_Rules.pdf #1141 (RIP)	Specification	Granted	
OBC_2023_9.8_annotated_clauses_40e96ca9.txt #1491 (RIP)	Specification	Granted	
OBC_2023_9.8_annotated_clauses_66ab9356.txt #1545 (RIP)	Specification	Granted	
OBC_2023_9.8_annotated_clauses.txt #1347 (RIP)	Specification	Granted	
SBC-201-2007.docling.dclg #1184 (RIP)	Specification	Granted	
SBC-201-2007.pdf #1185 (RIP)	Specification	Granted	

Figure 31: Document Access